EXPERIMENT NO 18

Determination and plotting of Antenna Half power beamwidth.

Aim:

To determine and plot Half Power Beamwidth (HPBW) and to determine the angular width at -3 dB points

Software required:

- SCILAB version 6.0.1

Program:


EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

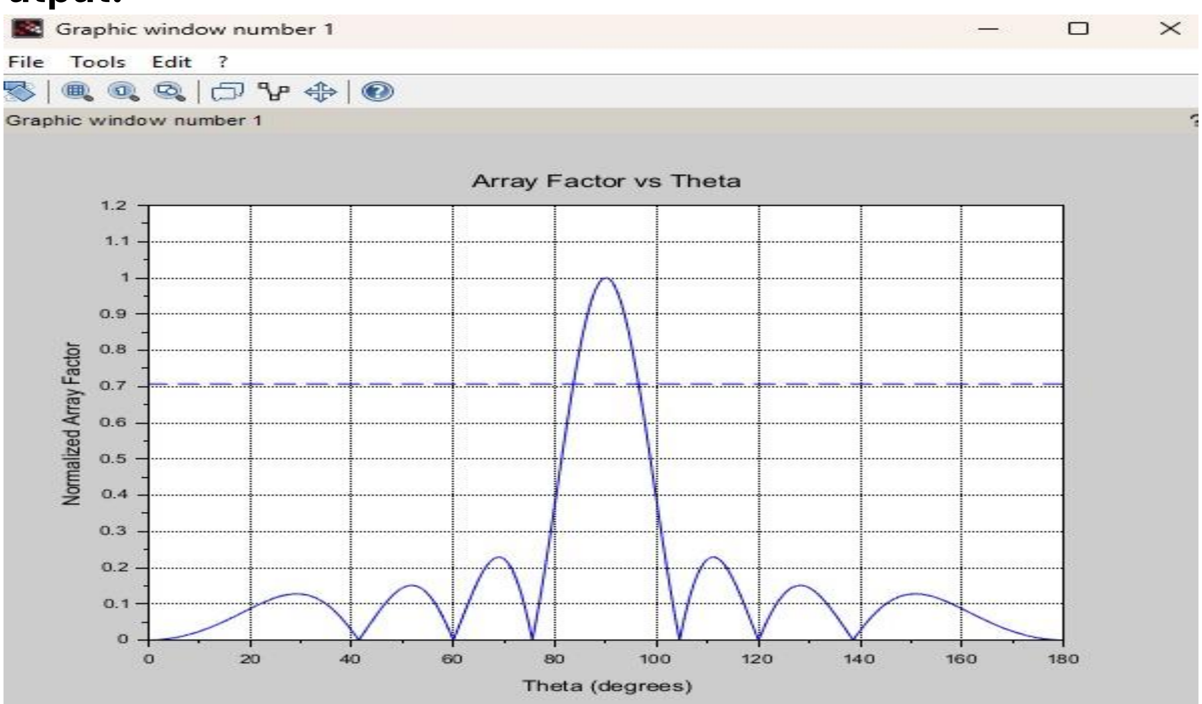
EXP 14.sci

```
clc;
clear;
clf();

// ==================================================
// ANTENNA ARRAY PARAMETERS
// ==================================================

N = 8;                  // Number of elements
d = 0.5;                // Element spacing in wavelengths
beta = 0;               // Progressive phase shift

// Angular range: 0 to 180 degrees
theta_deg = linspace(0, 180, 2000);
theta = theta_deg * %pi / 180;


// ==================================================
// ARRAY FACTOR
// ==================================================

// For d = lambda/2, kd = pi
psi = %pi * cos(theta) + beta;

AF = zeros(1, length(theta));

for i = 1:length(theta)

    denominator = sin(psi(i) / 2);

    if abs(denominator) < 1e-10 then
        AF(i) = 1;
    else
        AF(i) = abs(sin(N * psi(i) / 2) / ...
                    (N * denominator));
    end

end


// ==================================================
```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

```
end


// ================================================
// DISPLAY HPBW
// ================================================

mprintf("\n========================================\n");
mprintf("     HALF POWER BEAMWIDTH (HPBW)\n");
mprintf("========================================\n");
mprintf("Number of Elements = %d\n", N);
mprintf("Element Spacing    = %.2f lambda\n", d);
mprintf("HPBW               = %.3f degrees\n", HPBW);
mprintf("========================================\n");


// ================================================
// PLOT ARRAY FACTOR
// ================================================

figure(1);

plot(theta_deg, AF);

xlabel("Theta (degrees)");
ylabel("Normalized Array Factor");
title("Array Factor vs Theta");

xgrid();

a = gca();
a.data_bounds = [0, 0; 180, 1.05];


// ================================================
// SHOW HALF-POWER LEVEL
// ================================================

plot(theta_deg, threshold * ones(theta_deg), "--");

```

Output:



RESULT: The experiment was done and the Half power beamwidth of the given antenna was calculated. Different case studies were done.